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# Test-Time Perturbation Tuning with Delayed Feedback for Vision-Language-Action Models

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## Abstract

*Vision-Language-Action models (VLAs) achieve remarkable performance in sequential decision-making but remain fragile to subtle environmental shifts, such as small changes in object pose. We attribute this brittleness to trajectory overfitting, where VLAs over-attend to the spurious correlation between actions and entities, then reproduce memorized action patterns. We propose Perturbation learning with Delayed Feedback (PDF), a verifier-free test-time adaptation framework that improves decision performance without fine-tuning the base model. PDF mitigates the spurious correlation through uncertainty-based data augmentation and action voting, while an adaptive scheduler allocates augmentation budgets to balance performance and efficiency. To further improve stability, PDF learns a lightweight perturbation module that retrospectively adjusts action logits guided by delayed feedback, correcting overconfidence issue. Experiments on LIBERO (+7.4% success rate) and Atari (+10.3 human normalized score) demonstrate consistent gains of PDF in task success over vanilla VLA and VLA with test-time adaptation, establishing a practical path toward reliable test-time adaptation in multimodal decision-making agents. The code is available at https://github.com/zhoujiahuan1991/CVPR2026-PDF.*

## 1. Introduction

Vision-Language-Action models (VLAs) [13, 18, 23, 30, 31] have emerged as a powerful paradigm for sequential decision-making, demonstrating strong performance in robotic manipulation [2, 11, 12] and visually grounded control tasks [7, 24]. By jointly modeling visual, language, and action, VLAs translate instructions into executable actions through multimodal pretraining, marking a promising step toward general-purpose embodied intelligence.

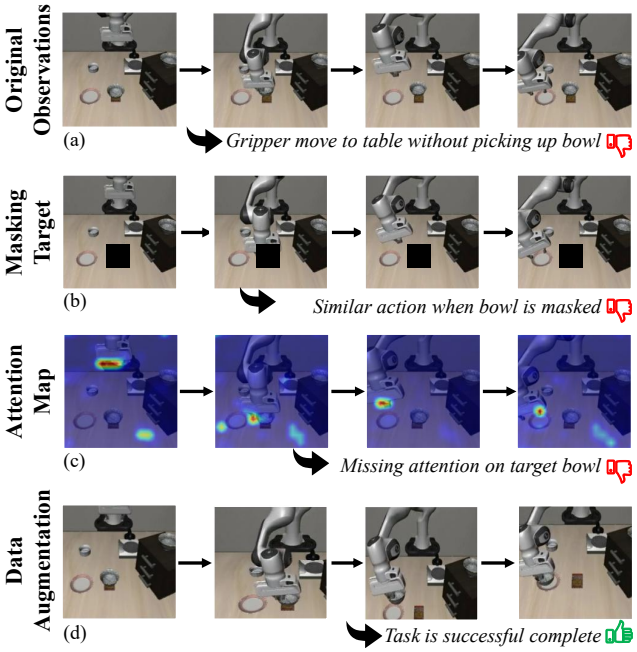



Figure 1. Evidence of trajectory overfitting and the effectiveness of data augmentation. In the first row, the gripper imitates expert trajectories regardless of task success. In the second row, it still reproduces similar actions when the target is masked. The third row shows that the gripper overlooks the target in attention maps. In contrast, after data augmentation, the gripper refocuses on the target and makes correct decisions.

Despite these impressive capabilities, VLAs remain fragile under even minor, semantically insignificant environmental shifts, such as slight changes in object pose, of-

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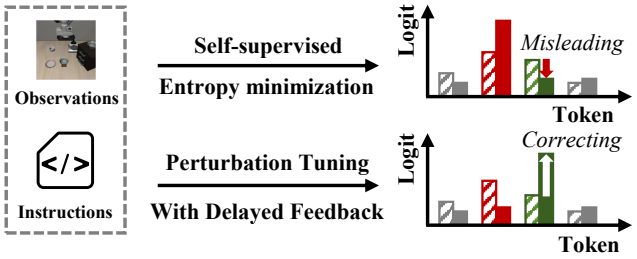



Figure 2. Comparison between traditional self-supervised test-time adaptation (TTA) and our PDF. Dashed bars indicate logits before adaptation; solid bars indicate logits after adaptation. Red bars denote incorrect-action logits; green bars denote correct-action logits. When VLAs become overconfident in incorrect behaviors, entropy-minimization-based TTA can amplify these errors by further boosting the wrong logits. In contrast, PDF corrects such misbehaviors by jointly mitigating the overconfidence issue and elevating the logits of correct actions, guiding the model toward accurate decisions.

ten leading to severe performance degradation or task failure [6]. This brittleness reveals a gap in test-time decision performance. To further elucidate this issue, we conduct diagnostic experiments that expose an intriguing behavioral bias. As shown in Figure 1(a), when instructed to grasp a bowl, the robotic arm may fail to complete the action but continues moving toward the plate, suggesting an overreliance on learned spatial or visual correlations rather than true semantic understanding. Even when the target bowl is masked (Figure 1(b)), the agent exhibits similar motions, indicating that the policy is not grounded in the visual semantics of the goal. We refer to this failure mode as trajectory overfitting, where VLAs memorize trajectory-specific visual or contextual patterns—such as gripper appearance or background textures—associated with success during training (Figure 1(c)), and reproduce suboptimal action traces when inputs resemble past trajectories. As shown in Figure 1(d), applying controlled data augmentation to pixel observations can restore correct execution, suggesting that perturbing the input distribution helps mitigate trajectory overfitting. This motivates improving decision performance via perturbation learning without costly fine-tuning.

One promising avenue to mitigate trajectory overfitting in VLAs is test-time adaptation (TTA) [29, 33], which allows models to refine their policies dynamically during inference. Among existing approaches, the most widely adopted are verifier-based TTA methods [14, 22], which employ a pretrained verifier to score candidate actions and select the most promising one via a best-of-$N$ voting scheme. Although effective, these methods incur substantial computational overhead from verifier pretraining and repeated rollouts, and they struggle to generalize to unseen environments with different dynamics or visual statistics. To address these limitations, recent efforts have explored verifier-free TTA strategies for VLAs [10], which adapt models in real time using only unlabeled test samples. However, without ground-truth feedback, these methods often suffer from adaptation instability or suboptimal accuracy under severe distribution shifts [15]. The main limitation lies in their reliance on self-supervised confidence metrics—such as entropy minimization [27]—which may become *unreliable* when model predictions are miscalibrated.

To overcome these limitations, we propose Perturbation learning with Delayed Feedback (PDF), a lightweight and verifier-free TTA framework designed to improve VLA robustness without parameter updates, which is demonstrated in Figure 2. PDF consists of two key components: (1) Uncertainty-Based Action Voting, which adaptively allocates augmentation resources based on the model's decision uncertainty, efficiently balancing decision performance and inference cost; and (2) Delayed Feedback-Guided Adaptation, a logit perturbation module with delayed feedback, which refines the model's action distribution using delayed feedback signals from the environment to correct overconfident decisions in a TTA manner. Extensive experiments across multiple VLA benchmarks, including robotic manipulation LIBERO and visual control games Atari, demonstrate that PDF substantially improves test-time decision performance while maintaining real-time efficiency.

Our main contributions are summarized as follows:

1. We systematically identify and analyze the phenomenon of trajectory overfitting, where VLAs rely on the spurious correlation between actions and entities, instead of semantic task grounding, revealing a fundamental cause of their fragility under environmental shifts.
2. We propose PDF, a verifier-free TTA framework that enhances VLA decision performance through uncertainty-based action voting and delayed feedback-guided adaptation, while keeping the base model parameters frozen.
3. Extensive experiments across diverse VLA benchmarks, including LIBERO and Atari, demonstrate that PDF consistently improves test-time decision performance.

## 2. Related Works

### 2.1. Vision-Language-Action Models

The strong open-world perception and reasoning capabilities of Vision-Language Models (VLMs) have driven their rapid adoption in robotic control tasks. Recently, Vision-Language-Action models (VLAs) [9, 16, 19, 25, 26, 30, 32] have emerged as an integrated and promising approach for bridging perception, language, and control. OpenVLA [11], a prominent open-source framework, supports training VLA models by integrating vision, language, and robotic control, providing a scalable platform for embodied-AI research. Gato [24], developed by DeepMind, is a generalist model that unifies vision, language, and action tasks

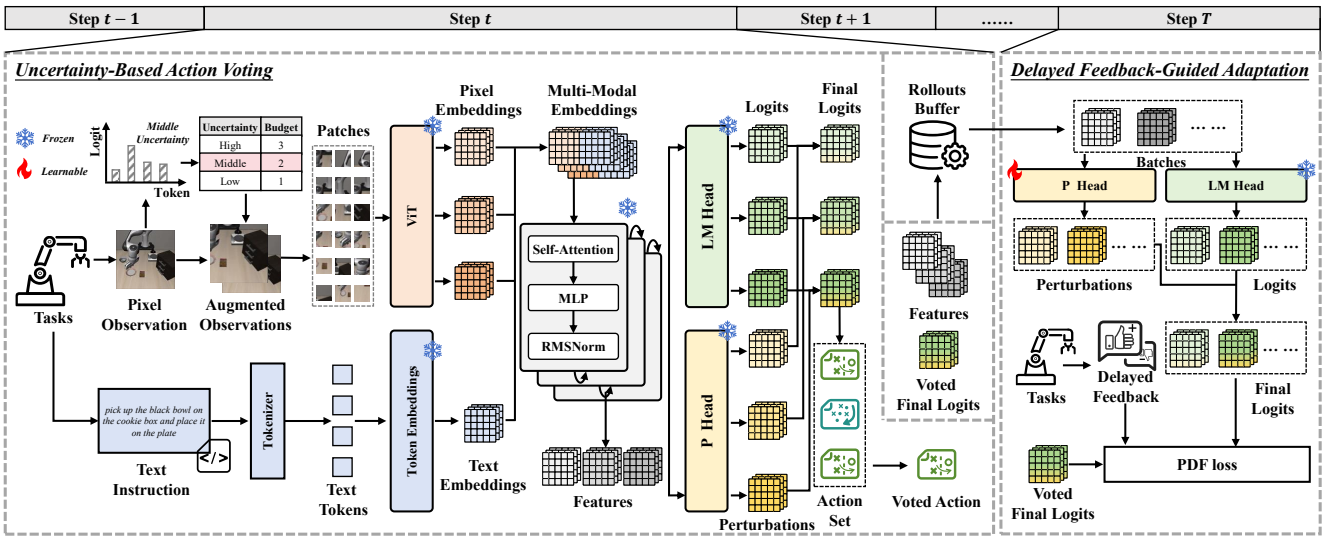



Figure 3. The overall framework of our proposed PDF. At test time, the VLA receives pixel observation $o_t$ and instruction $c_t$. Action-logit uncertainty $\mathcal{U}_t$ is estimated to allocate an adaptive augmentation budget $N_t$. Original and augmented observations are jointly encoded into multimodal features $f_t$, which are passed to an LM head to produce decision logits and to a perturbation head $h_\theta(\cdot)$ to generate logit perturbations. Final logits yield candidate actions, and the final action $a_t$ is selected by majority voting. The features and the voted final logits are stored in a rollout buffer $\mathcal{D}$. After each episode, a batch of features, voted final logits, and feedback is sampled to compute the PDF loss (Eq. 5). Only the P head $h_\theta(\cdot)$ is updated, enabling efficient and stable adaptation while keeping all VLA parameters frozen.

within a single architecture and demonstrates strong versatility across diverse modalities. RT-1 [34] and RT-2 [34], developed by Google, are robotic transformers for real-world manipulation tasks, with RT-2 extending RT-1 by incorporating language understanding to support more complex instruction following. Collectively, these works advance the integration of vision, language, and action in autonomous robotic agents.

### 2.2. Test-Time Adaptation in Vision-Language-Action Models

To improve the robustness and adaptability of VLAs in real-world settings, TTA has been actively explored. FEEDTTA [13] performs online adaptation for vision-language navigation via reinforcement learning with binary episodic feedback, but on a small navigation model. V-GPS [22] reranks action candidates from a pretrained VLA using a learned value function, enabling test-time policy selection without fine-tuning. RoboMonkey [14] scales at test time via sampling-and-verification with a VLM-based verifier. By contrast, SimpleVLA-RL [17] scales during training—rather than at test time—using reinforcement learning with outcome rewards. MG-Select [10] estimates confidence via KL divergence between predicted and reference action distributions and selects the best action among samples without an external verifier.

Our PDF instead improves confidence by mitigating the spurious correlation through data augmentation and refining the policy with delayed feedback, yielding a robust TTA scheme for VLAs.

## 3. Methods

In this section, we present Perturbation learning with Delayed Feedback (PDF), a plug-and-play TTA strategy designed for VLAs. Figure 3 depicts the overall architecture of our proposed PDF. The following subsections elaborate on each core component.

### 3.1. Prelimiaries

We begin by formulating the VLAs problem as a Markov Decision Process (MDP), defined by the tuple $M = (\mathcal{S}, \mathcal{A}, P, R)$, where $\mathcal{S} \subseteq \mathbb{R}^n$ and $\mathcal{A} \subseteq \mathbb{R}^m$ denote continuous state and action spaces, respectively. At each time step $t$, the agent observes a multimodal state $s_t = (o_t, c_t) \in \mathcal{S}$, consisting of a pixel observation $o_t \in \mathcal{O}$ and a textual instruction $c_t \in \mathcal{C}$. Given the current state $s_t$, the agent selects an action $a_t \in \mathcal{A}$, causing the environment to transition to a new state $s_{t+1} \sim P(\cdot \mid s_t, a_t)$. Upon task completion, the agent receives delayed feedback signals $r_i \in \mathcal{R}, i \in [1, T]$, which encapsulate task completion status or reward information.

### 3.2. Uncertainty-Based Action Voting

To improve the robustness of VLAs during test time, when VLAs interact with environments, the uncertainty of the model's decision is quantified using uncertainty estimation. Based on the estimated uncertainty, adaptive data augmen-

tation is applied to mitigate overfitting and attention imbalance, thereby improving the reliability of action generation while preserving inference efficiency.

Specifically, given a state $s_t = (o_t, c_t)$, the pixel observation $o_t$ is encoded by a visual encoder into pixel embeddings $e_{o_t}$, while the textual instruction $c_t$ is tokenized and embedded into text embeddings $e_{c_t}$. Then, both embeddings are concatenated and projected into features $f_t$. At last, the features are processed by an MLP $h_\phi(\cdot)$, referred to as the Large Model head (LM head), to generate the logits $z_t$ for each token.

To evaluate the uncertainty of the model's decision under the current state, we estimate predictive uncertainty from the output logits, defined as the normalized Shannon entropy of the predicted action distribution:

$$\mathcal{U}_t = -\frac{1}{\log K} \sum_{k=1}^{K} p(a_k \mid s_k) \log p(a_k \mid s_k), \tag{1}$$

where $K$ denotes the number of possible action tokens and $p(a_t = a_k \mid s_t) = \mathrm{softmax}(z_t)_k$.

In sequential decision-making tasks, a single erroneous action can result in complete task failure, with high-uncertainty actions being particularly prone to causing such outcomes. To mitigate action-generation uncertainty, we employ a simple yet effective strategy—data augmentation. However, data augmentation introduces significant computational overhead and therefore should be applied selectively. Therefore, augmented observations are generated by applying a set of transformations $\mathcal{T} = \{T_1, \ldots, T_N\}$, where the augmentation budget is adaptively determined by the model's uncertainty:

$$N_t = N_{\max} \cdot \mathcal{U}_t, \tag{2}$$

where $N_{\max}$ denotes the maximum augmentation budget. Both the original observation $o_t$ and its augmented views $\{T_j(o_t)\}_{j=1}^{N_t}$ are then divided into patches and encoded into pixel embeddings. The resulting pixel embeddings are concatenated with text embeddings to form a unified multimodal sequence and processed by the causal transformer to produce features $f_t$.

Finally, these features are projected by two parallel heads, the LM head $h_\phi(\cdot)$ and the Perturbation head (P head) $h_\theta(\cdot)$:

$$\tilde{z}_t = h_\phi(f_t) + \lambda h_\theta(f_t), \tag{3}$$

where $h_\phi(\cdot)$ generates the decision logits, $h_\theta(\cdot)$ outputs the learnable perturbations and $\lambda$ is a hyper-parameter. The resulting final logits $\tilde{z}_t$ are detokenized into a set of candidate actions, from which the final action $a_t$ is then selected through majority voting. The features of both the original and augmented observations are stored in a rollout buffer $\mathcal{D}$ for optimization once feedback is available.

It is important to note that all parameters of the VLA-including the visual encoder, token embedding layers, causal transformer, LM head and P head-remain frozen throughout this process.

### 3.3. Delayed Feedback-Guided Adaptation

Upon an episode completion, the agent receives delayed feedback $r \in \mathcal{R}$, which reflects overall task performance, such as task success, failure, or cumulative reward. A batch of features $f_b$ is then sampled from the rollout buffer $\mathcal{D}$, and the final logits $\tilde{z}_b$ with gradients are computed as

$$\tilde{z}_b = h_\phi(f_b) + \lambda h_\theta(f_b). \tag{4}$$

Then, the policy $\tilde{\pi}_b$ is generated by applying the softmax function to the final logits $\tilde{z}_b$. And the P head is subsequently optimized by

$$\mathcal{L}_{\mathrm{PDF}} = -(r - b) \log \pi_\phi + \lambda_{\mathrm{KL}} \mathbb{I}[r > b] \, \mathrm{KL}(\pi_\phi \parallel \tilde{\pi}), \tag{5}$$

where the first term is a REINFORCE-style objective increasing the likelihood of actions from the perturbed policy $\tilde{\pi}(s_t)$ when feedback $r$ exceeds the baseline $b$. A KL regularizer is gated by $\mathbb{I}[r > b]$, applied only under positive feedback to stabilize updates and accelerate convergence while allowing flexible exploration when $r \leq b$. Perturbations are thus steered toward successful action directions and discouraged otherwise. Only the P head $h_\theta$ is updated; all other VLA parameters remain frozen for efficient, stable test-time adaptation.

## 4. Experiments

To evaluate the effectiveness of the proposed PDF, we conduct experiments across diverse decision-making simulation environments. Specifically, we employ OpenVLA [11], a VLA fine-tuned on the LIBERO [20] benchmark and Jat [7]—an open-source implementation of GATO [24]—for the Atari suite [1]. This combination of environments and baselines enables a comprehensive evaluation of the superiority of our method across diverse task domains.

### 4.1. Experimental Setup

LIBERO [20]: We use four 10-task suites (Spatial, Object, Goal, Long) in the LIBERO Franka Panda simulation with RGB views, robot states, task text, and delta end-effector actions. Each task: 50 eval rollouts; metric is success rate. Policies are trained for 50 episodes with OpenVLA [11] before evaluation.

Atari [1]: We evaluate all 57 Atari 2600 games (pixel input, 4–18 discrete actions). Episodes end on life loss, level completion, or at 108,000 frames. Each game: 50 eval episodes; metric is Human-Normalized Score ($HNS = (\mathrm{Score} - \mathrm{RandomScore})/(\mathrm{HumanScore} - \mathrm{RandomScore})$). Base model Jat [7] is trained 50 episodes

Table 1. Performance comparison on the LIBERO benchmark (Spatial, Object, Goal, Long). † denotes reproduced results and SR indicates Success Rate (%). Blue numbers indicate SOTA within each task suite.

| Method | Pub. | Param. | Spatial | | Object | | Goal | | Long | | Avg. | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | SR ↑ | Rank ↓ | SR ↑ | Rank ↓ | SR ↑ | Rank ↓ | SR ↑ | Rank ↓ | SR ↑ | Rank ↓ |
| PackNet [21] | CVPR'18 | - | 0.63 | 11 | 0.60 | 10 | 0.75 | 8 | 0.25 | 12 | 0.56 | 10.2 |
| ER [3] | Arxiv'19 | - | 0.56 | 12 | 0.44 | 12 | 0.49 | 11 | 0.32 | 11 | 0.45 | 11.5 |
| SeqL [20] | NeurIPS'23 | - | 0.20 | 13 | 0.26 | 13 | 0.22 | 12 | 0.15 | 13 | 0.21 | 12.8 |
| MTL [20] | NeurIPS'23 | - | 0.83 | 6 | 0.54 | 11 | 0.80 | 3 | 0.48 | 9 | 0.66 | 7.2 |
| ATM [28] | RSS'23 | - | 0.69 | 10 | 0.68 | 8 | 0.78 | 5 | 0.39 | 10 | 0.63 | 10.5 |
| OpenVLA [11] | CoRL'24 | - | 0.79 | 8 | 0.86 | 4 | 0.85 | 2 | 0.51 | 7 | 0.75 | 5.2 |
| OpenVLA† [11] | CoRL'24 | - | 0.85 | 2 | 0.64 | 9 | 0.76 | 6 | 0.53 | 6 | 0.69 | 5.8 |
| DP [5] | IJRR'25 | - | 0.78 | 9 | **0.92** | **1** | 0.68 | 10 | 0.51 | 8 | 0.72 | 7 |
| OCTO [8] | RSS'24 | 93M | 0.79 | 8 | 0.86 | 4 | 0.85 | 2 | 0.51 | 7 | 0.75 | 5.3 |
| TraceVLA [32] | ICLR'25 | 130M | 0.85 | 4 | 0.85 | 5 | 0.75 | 7 | 0.54 | 4 | 0.75 | 6.5 |
| OpenVLA-DPO [4] | Arxiv'25 | 130M | 0.84 | 5 | 0.89 | 2 | 0.79 | 4 | 0.53 | 5 | 0.76 | 4 |
| SFT-4LIBERO [16] | Arxiv'25 | 130M | 0.85 | 3 | 0.87 | 3 | 0.77 | 5 | 0.55 | 3 | 0.76 | 3.5 |
| MG-Select [10] | Arxiv'25 | 130M | 0.82 | 7 | 0.73 | 6 | 0.73 | 9 | 0.55 | 2 | 0.71 | 6 |
| **PDF (Ours)** | - | 9M | **0.90** | **1** | 0.72 | 7 | **0.86** | **1** | **0.59** | **1** | **0.77** | **2.5** |

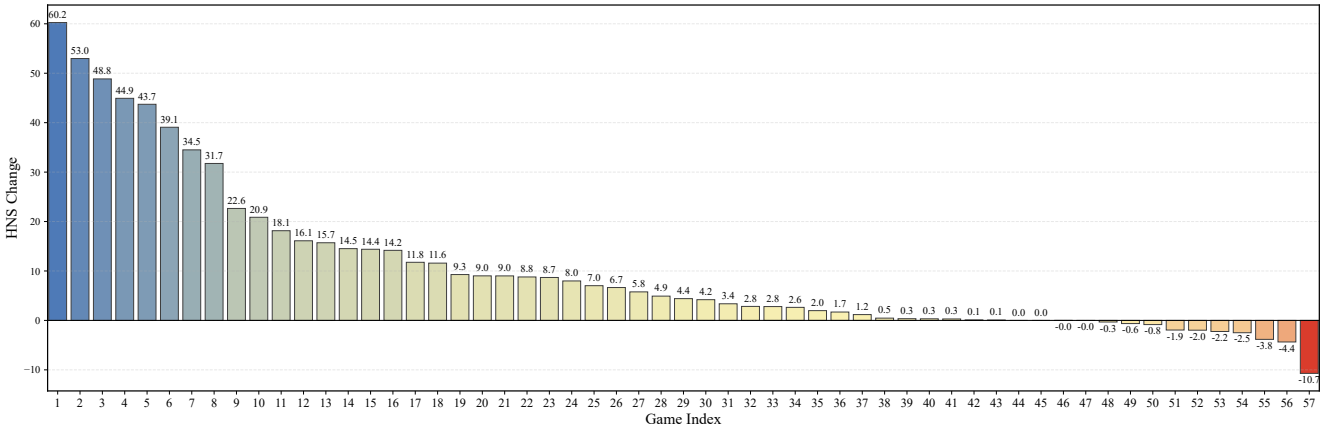



Figure 4. Human normalized score changes across 57 Atari games. Blue bars show performance improvements, orange bars indicate degradation. Games are sorted by improvement magnitude, with BOXING (+60.25%) showing maximum gain and BATTLE ZONE (-10.72%) showing maximum decline. 47/57 games demonstrate positive performance changes, with 11.28% mean improvement.

per game. All LIBERO and Atari experiments run on a single Tesla V100-PCIE-32GB.

## 4.2. Performance Comparison Across Benchmarks

To demonstrate the effectiveness of PDF, we conduct extensive experiments across multiple benchmarks.

Table 1 summarizes results on four LIBERO suites. PDF attains the best average performance (0.77 SR, 2.5 mean rank), exceeding prior VLA variants and adaptation baselines. It reaches 0.89 on Spatial (vs. 0.85 for reproduced OpenVLA) and 0.86 on Goal, slightly ahead of OCTO and OpenVLA. Gains are largest on Long-horizon tasks (0.59, +4.1 over the best baseline), indicating stronger sequential decision quality. Compared to MG-Select, PDF improves average SR by nearly 6 points, showing the advantage of uncertainty-weighted action voting and delayed-feedback perturbation learning over sample filtering. PDF also matches or surpasses strong pretrained models (TraceVLA, SFT-4LIBERO) without any fine-tuning. These results underscore PDF's effectiveness in mitigating trajectory overfitting and boosting robustness in spatially complex, long-horizon settings.

PDF is also evaluated on Atari-57, attaining 1.07 HNS vs. 0.97 for the Jat base (+0.10). Figure 4 shows per-game HNS deltas: 47/57 games improve. Largest gains appear in BOXING (+0.60), TIME PILOT (+0.53), ATLANTIS (+0.49), and ASSAULT (+0.45). Our method uses 9M trainable parameters and requires no gradient access to the base model, whereas baselines use 93–130M parameters and require base-model gradients.

### 4.3. Impact of Data Augmentation and Feedbacks

To evaluate the impact of the data augmentation (DA) mechanism and perturbation learning with delayed feedback (DF), we conduct extensive experiments on the four LIBERO suites. Note that "PDF w/o DF" denotes OpenVLA that selects actions solely via multi-view voting (without delayed feedback), and "PDF w/o DA" denotes OpenVLA that learns perturbations using only the original pixel observations (without data augmentation). Table 2 presents an ablation comparing the baseline OpenVLA with variants of the proposed PDF across the LIBERO suites. Results show that integrating both DF and DA in PDF is effective. Overall, PDF attains the highest average success rates across all four suites. This superiority holds for most tasks within each suite, as indicated by boldface entries in the table.

Moreover, the ablation reveals the contribution of each component. The PDF w/o DA variant surpasses the OpenVLA baseline on most tasks. By contrast, the PDF w/o DF variant shows a substantial drop across suites, with success rates dropping to 0.50 on Object and 0.77 on Goal. These results suggest that DF is critical for robustness, particularly in object-manipulation and goal-conditioned tasks. The synergy between DF and DA is evident. The full PDF model outperforms both variants and yields more stable results across suites, confirming that both DF and DA are integral to the observed performance gains.

### 4.4. Ablation Study on Data Augmentation Budget

Given that data augmentation (DA) is a crucial module in PDF and its budget affects performance, we evaluate its impact across the four LIBERO suites and the Atari benchmark. Figure 5 summarizes performance under five augmentation budgets (0-4, higher uncertainty indicates more data augmentations) for the LIBERO and Atari benchmarks. As the test-time augmentation budget (i.e., the number of augmented views sampled for voting) increases, performance generally declines across benchmarks. Increasing the data augmentation budget in PDF generally degrades performance across LIBERO and Atari benchmarks due to noise accumulation, with a small budget (max of 3) offering the best trade-off to mitigate trajectory overfitting without introducing low-quality views. Adaptive, benchmark-specific budget ceilings are recommended

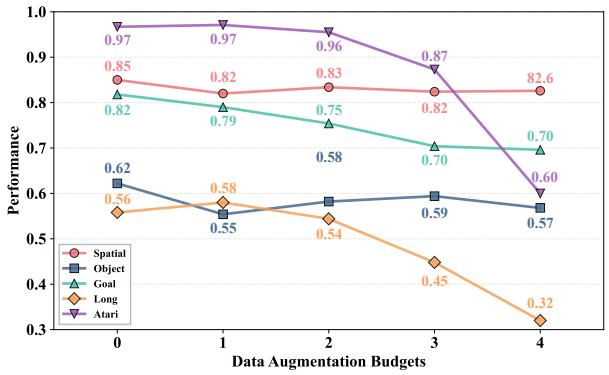



Figure 5. Performance degradation under increasing data augmentation budgets. All five benchmarks (LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, LIBERO-Long, and Atari) show performance decline as augmentation budget increases from 0 to 4, indicating that excessive data augmentation harms model effectiveness.

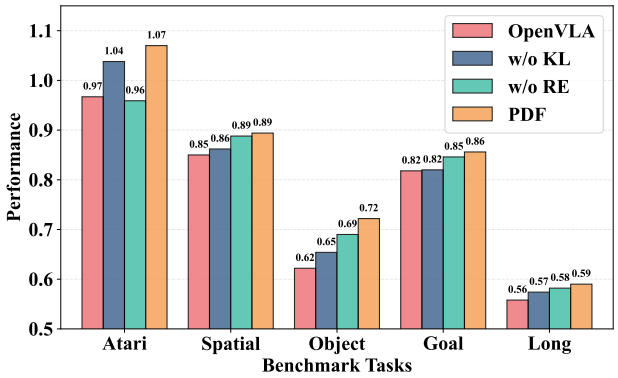



Figure 6. Performance comparison across five benchmarks shows that both the KL divergence and the REINFORCE-style term contribute to the full PDF model's superior performance. Removing either component results in degraded performance, with KL divergence showing greater impact on final results.

for optimal results in structure-dependent tasks. Overall, we set the max data augmentation budget to 3, because moderate augmentation is benign, whereas aggressive budgets degrade structure-dependent tasks; adaptive, benchmark-specific ceilings are preferable to uniform maximization.

### 4.5. Ablation Study on Loss Functions

The PDF objective comprises two components: a REINFORCE-style policy gradient term and a KL regularization term. To evaluate the contribution of each term, we conduct experiments on the four LIBERO suites and the Atari benchmark. Figure 6 compares: (i) the base model OpenVLA, (ii) a variant without the KL term (w/o KL), (iii) a variant without the REINFORCE-style term

Table 2. Performance comparison of OpenVLA and PDF variants (with and without Delayed Feedback (DF) and Data Augmentation (DA)) across LIBERO benchmark suites (Spatial, Object, Goal, and Long). Reported as success rates over 10 tasks with average performance.

| Suite | Method | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | Avg. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Spatial** | OpenVLA | 0.94 | 0.94 | 0.86 | 0.96 | 0.64 | 0.88 | 0.92 | 0.86 | 0.74 | 0.76 | 0.85 |
| | PDF w/o DF | 0.78 | 0.84 | 0.90 | 0.90 | 0.78 | 0.68 | 0.92 | 0.70 | 0.80 | 0.74 | 0.80 |
| | PDF w/o DA | 0.94 | 0.94 | 0.88 | 0.94 | 0.68 | 0.88 | 0.92 | 0.88 | 0.76 | 0.76 | 0.86 |
| | PDF | 0.94 | 0.96 | 0.98 | 0.98 | 0.80 | 0.88 | 0.92 | 0.88 | 0.80 | 0.80 | 0.89 |
| | $\Delta \uparrow$ | 0.00 | 0.02 | 0.12 | 0.02 | 0.16 | 0.00 | 0.00 | 0.02 | 0.06 | 0.04 | 0.04 |
| **Object** | OpenVLA | 0.64 | 0.68 | 0.78 | 0.34 | 0.90 | 0.66 | 0.50 | 0.62 | 0.36 | 0.74 | 0.62 |
| | PDF w/o DF | 0.86 | 0.50 | 0.74 | 0.62 | 0.68 | 0.16 | 0.10 | 0.28 | 0.36 | 0.74 | 0.50 |
| | PDF w/o DA | 0.88 | 0.68 | 0.82 | 0.40 | 0.94 | 0.70 | 0.48 | 0.66 | 0.36 | 0.76 | 0.67 |
| | PDF | 0.96 | 0.74 | 0.82 | 0.66 | 0.96 | 0.66 | 0.56 | 0.66 | 0.40 | 0.80 | 0.72 |
| | $\Delta \uparrow$ | 0.32 | 0.06 | 0.04 | 0.32 | 0.06 | 0.00 | 0.06 | 0.04 | 0.04 | 0.06 | 0.10 |
| **Goal** | OpenVLA | 0.62 | 0.90 | 0.84 | 0.66 | 0.94 | 0.90 | 0.78 | 1.00 | 0.86 | 0.68 | 0.81 |
| | PDF w/o DF | 0.70 | 0.96 | 0.88 | 0.34 | 0.94 | 0.78 | 0.82 | 0.98 | 0.68 | 0.62 | 0.77 |
| | PDF w/o DA | 0.62 | 0.94 | 0.86 | 0.60 | 0.92 | 0.94 | 0.80 | 1.00 | 0.76 | 0.66 | 0.81 |
| | PDF | 0.72 | 0.98 | 0.90 | 0.66 | 0.96 | 0.96 | 0.86 | 1.00 | 0.82 | 0.70 | 0.85 |
| | $\Delta \uparrow$ | 0.10 | 0.08 | 0.06 | 0.00 | 0.02 | 0.06 | 0.08 | 0.00 | -0.04 | 0.02 | 0.04 |
| **Long** | OpenVLA | 0.62 | 0.64 | 0.58 | 0.50 | 0.44 | 0.86 | 0.50 | 0.66 | 0.20 | 0.58 | 0.56 |
| | PDF w/o DF | 0.70 | 0.64 | 0.58 | 0.54 | 0.52 | 0.82 | 0.48 | 0.62 | 0.20 | 0.52 | 0.56 |
| | PDF w/o DA | 0.72 | 0.60 | 0.58 | 0.50 | 0.46 | 0.88 | 0.50 | 0.66 | 0.20 | 0.56 | 0.57 |
| | PDF | 0.72 | 0.64 | 0.58 | 0.56 | 0.56 | 0.88 | 0.50 | 0.68 | 0.20 | 0.58 | 0.59 |
| | $\Delta \uparrow$ | 0.10 | 0.00 | 0.00 | 0.06 | 0.12 | 0.02 | 0.00 | 0.02 | 0.00 | 0.00 | 0.03 |

Table 3. Performance comparison of dim-wise voting and action-wise voting across Atari and LIBERO tasks.

| Task | Dim-wise | Action-wise |
|---|---|---|
| Alien | **0.26** | - |
| Black Bowl | 0.68 | **0.88** |
| Alphabet Soup | **0.86** | 0.64 |
| Bowl on Stove | **0.96** | 0.90 |
| Moka Pot | **0.58** | **0.58** |

(w/o RE), and (iv) the full PDF objective. The full objective yields the highest scores; the largest absolute/relative gains occur on Atari (+0.1) and LIBERO-Object (+0.1). Removing the KL term markedly boosts Atari (+0.07) but yields only modest gains on other suites; removing the REINFORCE-style term slightly harms Atari (-0.01) yet improves LIBERO-Object (+0.07) and shows smaller positive shifts elsewhere. Moderate improvements are observed on LIBERO-Spatial (+0.04) and LIBERO-Goal (+0.04), while LIBERO-Long shows a smaller gain (+0.03). These patterns suggest KL term and the REINFORCE-style term provide complementary benefits—stability and calibrated exploration vs. reward-aligned action refinement—whose combination yields additive, non-redundant gains.

### 4.6. Comparison between Voting Mechanisms

To evaluate voting strategies, we compare three variants. Dimension-wise voting takes a majority vote per action dimension, while action-wise voting votes over full action tuples. As shown in Table 3, on Atari (Alien) the action is effectively one-dimensional, making dimension-wise and action-wise voting equivalent. On LIBERO, dimension-wise voting best mitigates trajectory overfitting by allowing cross-view agreement on individual action components, enabling more flexible deviations from the original policy. Action-wise voting often falls back to the baseline because views rarely agree on the entire action tuple, yielding mixed results. Overall, dimension-wise voting provides the most reliable robustness–plasticity trade-off.

### 4.7. Visualization and Case Study

To further assess the effectiveness of PDF, we conduct qualitative case studies on three LIBERO tasks: (i) "pick up the cream cheese and place it in the basket" (*Pick up cream cheese*), (ii) "pick up the black bowl on the wooden cabinet and place it on the plate" (*Pick up black bowl*), and (iii) "put both the alphabet soup and the tomato sauce in the basket" (*Put alphabet soup*). As shown in Figure 7(a,b), OpenVLA exhibits trajectory overfitting: it executes a demonstration-like motion plan but disregards whether the target object

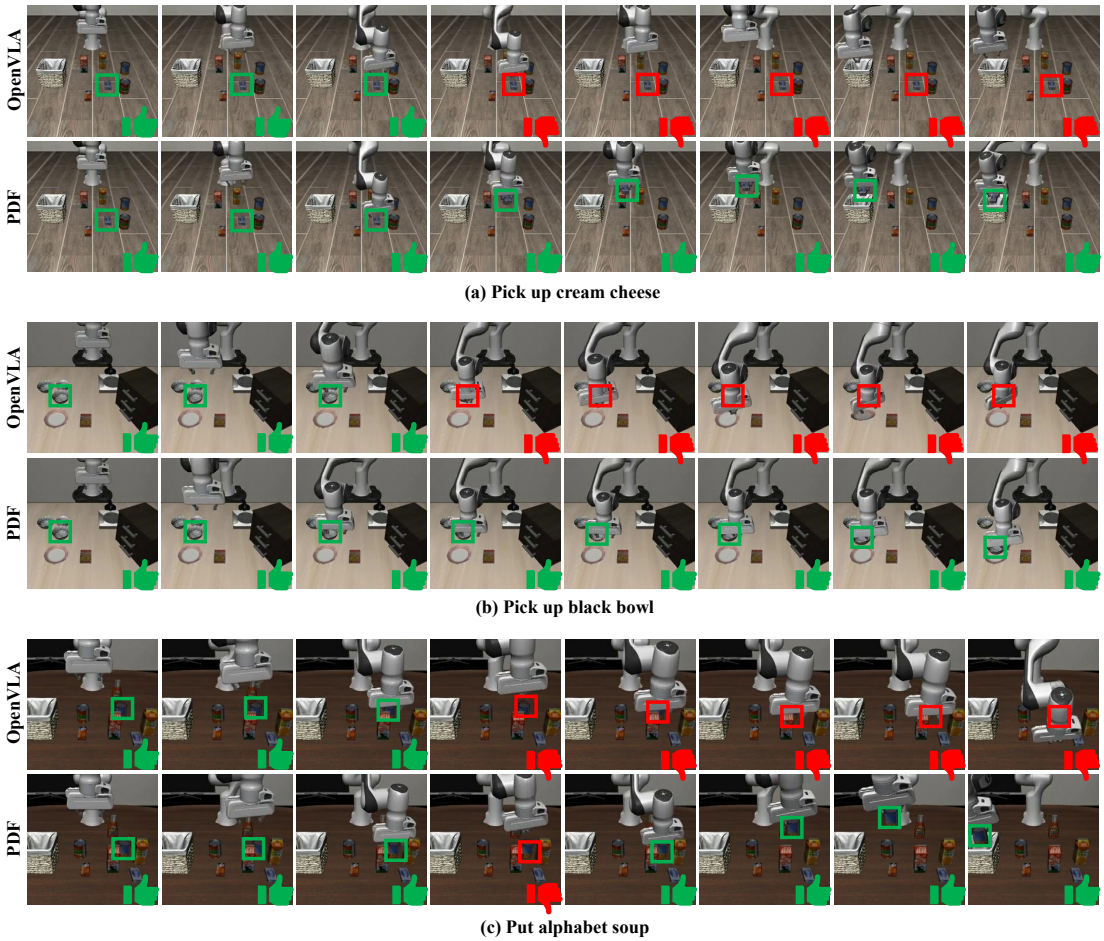



Figure 7. Visual comparison of OpenVLA and PDF on three tasks. The green thumb indicates that the agent performed the correct action, while the red thumb indicates an incorrect action. The box represents the target entity, with green indicating that the target entity has been operated correctly and red indicating incorrect operation.

has been successfully grasped, leading to failure to complete the tasks. In contrast, PDF manifests a more robust, state-aware policy under the same conditions and reliably completes both *Pick up cream cheese* and *Pick up black bowl*. Interestingly, on *Put alphabet soup* (Figure 7(c)), PDF initially fails to secure the alphabet soup but promptly reattempts and succeeds, indicating decision-making rather than open-loop replay. Overall, these case studies highlight PDF's resilience to demonstration bias and its capacity to leverage feedback for corrective actions, resulting in higher task completion reliability.

## 5. Conclusions and Limitations

We introduce PDF, a verifier-free TTA method that improves VLA decision performance in sequential decision making.VLAs suffer from trajectory overfitting—overweighting the spurious correlation and replaying memorized action chains—reducing reliability under subtle environmental shifts. PDF addresses this via (i) uncertainty-based action voting and (ii) delayed-feedback–guided adaptation. On LIBERO robotic manipulation and Atari-57, PDF surpasses vanilla VLAs and VLAs with TTA baselines, improving decision performance with minimal overhead. However, PDF does not fully solve trajectory overfitting; rather, it alleviates its negative impact. In future work, we will address the challenge more fundamentally.

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